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Modeling and forecasting intraday spot volatility☆

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ABSTRACT

We propose a multiple-equation regression-based method for modeling and forecasting intraday spot volatility. In this approach, intraday intervals are treated as individual time series, deviating from the common practice of treating the data as one continuous sample. Our empirical study, which spans more than two decades and encompasses six US blue-chip stocks, employs the recent OK volatility estimator developed by Li, Wang, and Zhang (2024) to expose the dynamics of latent intraday spot volatility over time. We demonstrate that the proposed method effectively captures the intricate dynamics of intraday spot volatility and find strong evidence that it outperforms a competing regression approach, and popular tree-based machine learning (LightGBM) and deep learning (LSTM) methods, in terms of predictive accuracy as measured by the MSE and QLIKE. These improvements in predictive accuracy extend to logarithmic extensions and across multiple forecast horizons. Overall, our results indicate that the parameter flexibility inherent in the proposed method is advantageous. This flexibility comes without undue computational burden.

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1. Introduction

This paper proposes a novel regression-based framework for forecasting intraday spot volatility. The approach is computationally inexpensive and simple to implement and interpret. It also provides for a direct link between overnight volatility and spot volatility during the trading day. Forecasts of volatility are important inputs into financial decisions in the context of risk management, portfolio construction, and derivative pricing, among others. While this has motivated a great deal of research on forecasting methods, the focus has been at a daily or lower frequency. Given the rapid growth in high-frequency trading (HFT), forecasts of intraday volatility are important for risk management, derivative pricing, market making, and informing HFT strategies; see for example (Engle & Sokalska, 2012), Stroud and Johannes (2014), and Bates (2019). While the need for intraday volatility forecasts has increased, relatively little attention (in comparison to daily volatility forecasts) has been paid to this issue. Taylor and Xu (1997) proposed an ARCH model for hourly exchange-rate volatility. Engle and Sokalska (2012) developed a component GARCH model containing diurnal, intraday, and daily components for volatility during the trading day. More recently, Zhang, Zhang, Cucuringu, and Qian (2023) applied a range of linear regression and machine learning methods to forecasting intraday realized volatility.

We propose a multiple-equation regression-based method for modeling and forecasting intraday spot volatility, where intraday intervals are treated as individual time series, as opposed to most methods that treat the data as

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one continuous sample. We employ the OK volatility estimator developed by Li, Wang, and Zhang (2024) to expose the dynamics of latent intraday spot volatility over time. By utilizing an observable measure of latent intraday spot volatility, regression methods can be employed instead of GARCH or other latent variable models. Our model specification is in the spirit of Engle and Sokalska (2012) in that diurnal, intraday, and longer-term daily effects are incorporated, the latter based on HAR-like structures. In contrast to Engle and Sokalska (2012) and Zhang et al. (2023), the proposed framework allows for a simple way to incorporate the effect of overnight volatility. This provides insights into how overnight volatility influences the evolution of spot volatility throughout the trading day, with the aim of improving forecasts. The proposed method is computationally inexpensive and allows for a clear and straightforward interpretation. Low computational cost is crucial in the high-frequency setting, as forecasts can then be generated on demand to inform HFT applications.

Our empirical study spans more than two decades and encompasses six US blue-chip stocks. Its in-sample results show that the estimated parameters within the proposed multiple-equation regression approach vary during the trading day, and that it adequately captures the persistence in intraday spot volatility. Its out-of-sample forecasting results show that the proposed method outperforms a recent regression-based alternative, as well as popular machine learning and deep learning methods, in terms of the MSE and QLIKE. The benefits of the proposed method are robust to logarithmic specifications, multi-step-ahead forecasts, the exclusion of overnight volatility, and to a number of simpler regression specifications. The study also shows that these benefits are not simply due to capturing the diurnal pattern in volatility, but rather to the proposed method's ability to capture more complex intraday dynamics.

The remainder of this paper is organized as follows. Section 2 describes the methodology, including spot volatility estimation, our proposed multiple-equation regression-based method, and the competing approaches. Section 3 reports the in-sample and out-of-sample forecasting results of the empirical study, along with a number of robustness checks, and Section 4 provides concluding remarks. Additional supporting results are available in the online appendix.

2. Methodology

This section describes the spot volatility estimator, our proposed multiple-equation regression-based framework, the benchmark models, and how forecast accuracy is assessed.

2.1. Spot volatility estimation

Motivated by the observation that the average investor lacks access to high-quality transaction data but has free access to intraday candlestick charts, Li et al. (2024) recently proposed the Optimal candlesticK (OK) estimator for spot volatility at a given time point. In a standard setting, they show that the OK estimator is asymptotically unbiased and has minimal asymptotic variance within a class of linear estimators. Their theoretical and numerical results suggest that the proposed candlestick-based estimator is much more accurate than the conventional spot volatility estimator based on high-frequency returns.

Begin by defining $O_{i,t}$, $H_{i,t}$, $L_{i,t}$, and $C_{i,t}$ as the logarithm of the open, high, low, and closing prices, respectively, in the ith intraday interval on trading day t. M denotes the number of intraday intervals per day. The return and range within an interval are given by $r_{i,t} = C_{i,t} - O_{i,t}$ and $w_{i,t} = H_{i,t} - L_{i,t}$. In its basic form, the non-parametric OK estimator of the spot volatility for interval i on day t is given by

$$OK_{i,t} = \frac{0.811 w_{i,t} - 0.369|r_{i,t}|}{\sqrt{\Delta}} \tag{1}$$

where Δ denotes the fixed length of the intervals.

As Li et al. (2024) argue, the above spot volatility estimator exhibits a form of optimality, provides reliable feasible inference, delivers sensible empirical estimates, and is very simple to compute. We use the OK estimator to estimate latent volatility within individual intraday intervals—specifically, to create predictors in competing spot volatility models and for spot volatility forecast comparisons.

2.2. A regression-based method for intraday spot volatility

We next describe the proposed regression-based method for modeling and forecasting intraday spot volatility. While not a component model, our specification is in the spirit of Engle and Sokalska (2012) in that diurnal, intraday, and longer-term daily effects are incorporated.

We adopt a multiple-equation approach, where each intraday interval $i = 1, 2, \ldots, M$ has its own linear equation, and hence its own set of parameters. As a result, the data for a financial asset are not treated as a single time series. Instead, each period of the day (intraday interval) has its own time series and is treated as a separate forecasting problem. This approach permits the overnight volatility to be directly incorporated in each equation, allowing for different effects across different periods within the trading day. In addition, period-specific intercept parameters capture the diurnal pattern in volatility. Multiple-equation approaches have been successfully applied across a wide range of forecasting applications, including electricity market load and air pollution; see Clements, Hurn, and Li (2016), Espinoza, Joye, Belmans, and De Moor (2005), Moisan, Herrera, and Clements (2018), Peirson and Henley (1994), Ramanathan, Engle, Granger, Vahid-Araghi, and Brace (1997), Soares and Medeiros (2008). These simple regression-based frameworks have been documented to perform as well as more complex machine learning methods.

In the proposed framework, the same linear specification is applied to all M intraday intervals. For ease of exposition, the model for intraday interval $i = 1$ is described first, followed by the specification for the remaining $M - 1$ intervals.

The model of the spot volatility for the first intraday interval on day t, given by

$$OK_{1,t} = \beta_{1,0} + \beta_{1,1}OK^{d}_{1,t-1} + \beta_{1,2}OK^{w}_{1,t-1} + \beta_{1,3}OK^{m}_{1,t-1} + \beta_{1,4}|r^{ON}_{t}| + \beta_{1,5}OK_{M,t-1} + \beta_{1,6}\overline{OK}^{5}_{M,t-1} + \beta_{1,7}\overline{OK}^{20}_{M,t-1} + \varepsilon_{1,t} \quad (2)$$

where

$$OK^{d}_{1,t-1} = OK_{1,t-1}, \quad OK^{w}_{1,t-1} = \tfrac{1}{5}\sum_{j=1}^{5} OK_{1,t-j}, \quad OK^{m}_{1,t-1} = \tfrac{1}{22}\sum_{j=1}^{22} OK_{1,t-j} \text{ and } r^{ON}_{t} = O_{1,t} - C_{M,t-1}$$

has (i) an interval specific intercept, (ii) a HAR-like component using the estimate of the spot volatility for the 1st intraday interval of the previous day and the average spot volatility estimate for this interval over the previous week and month as predictors, (iii) a predictor $|r^{ON}_{t}|$, and (iv) another HAR-like component using the estimated spot volatility for the Mth intraday interval of the previous day, along with the averages of the five and 20 most recent spot volatility estimates from the previous day as predictors. These intraday averages are given by

$$\overline{OK}^{5}_{M,t-1} = \tfrac{1}{5}\sum_{j=0}^{4} OK_{M-j,t-1} \text{ and } \overline{OK}^{20}_{M,t-1} = \tfrac{1}{20}\sum_{j=0}^{19} OK_{M-j,t-1}.$$

Motivated by the empirical success of the HAR model of Corsi (2009) for daily volatility, the first HAR-like specification in (2) aims to parsimoniously capture the strong daily persistence observed in the data. The second specification, while clearly not a standard HAR structure, aims to parsimoniously capture the observed strong intraday persistence. The impact of overnight volatility is captured by the parameter on $|r^{ON}_{t}|$.

The model of the spot volatility for the second to the Mth intraday interval on day t has a similar specification:

$$OK_{i,t} = \beta_{i,0} + \beta_{i,1}OK^{d}_{i,t-1} + \beta_{i,2}OK^{w}_{i,t-1} + \beta_{i,3}OK^{m}_{i,t-1} + \beta_{i,4}|r^{ON}_{t}| + \beta_{i,5}OK_{i-1,t} + \beta_{i,6}\overline{OK}^{5}_{i-1,t} + \beta_{i,7}\overline{OK}^{20}_{i-1,t} + \varepsilon_{i,t} \quad (3)$$

Each equation ($i = 2, \ldots, M$) has its own intercept, contains a daily component incorporating the estimate of the spot volatility for the ith intraday interval of the previous day and the average spot volatility estimate for this interval over the previous week and month, and includes the overnight volatility. Each equation also has an intraday component, incorporating the estimated spot volatility for the $(i-1)$-th intraday interval of day t and averages of the five and 20 most recent spot volatility estimates. Hence, these intraday averages are defined in the same way as the corresponding predictors in Eq. (2).

An alternative approach would be to increase M by 1 and use only $|r^{ON}_{t}|$ as a predictor of the spot volatility for the first intraday interval of day t. Models such as Eqs. (2) and (3), excluding the overnight volatility predictor, could then be applied to the remaining intraday intervals. While the results are not reported here, estimating such a model presents a problem, as it implies a nonstationary intraday effect for the first interval, which uses $|r^{ON}_{t}|$. This issue arises due to the regular jump in level from the volatility of the preceding trading day to the overnight volatility.

To assess the size of the jump in intraday volatility, we report summary statistics for two ratios: the relative magnitude of overnight volatility compared to (i) the volatility in the last interval of the preceding trading day, measured by $|r^{ON}_{t}|/OK_{M,t-1}$, and (ii) the volatility in the first interval of the succeeding trading day, measured by $|r^{ON}_{t}|/OK_{1,t}$. For IBM (one of the stocks considered in our empirical study), the empirical mean, median, 90th percentile, and the probability that the numerator exceeds the denominator for the ratio $|r^{ON}_{t}|/OK_{M,t-1}$ are 4.9472, 3.0673, 10.4536, and 0.7934, respectively. The corresponding estimates for $|r^{ON}_{t}|/OK_{1,t}$ are 2.2857, 1.3482, 5.1047, and 0.5955. These statistics indicate that overnight volatility is typically much larger than the volatility in the last interval of the preceding trading day. It is also generally larger than the volatility in the first interval of the succeeding trading day, although the estimates are lower due to the diurnal pattern in intraday volatility. The online appendix includes plots of both $|r^{ON}_{t}|/OK_{M,t-1}$ and $|r^{ON}_{t}|/OK_{1,t}$ for the IBM data, as well as the corresponding summary statistics for the other five stocks in our empirical study. Overall, the results suggest that overnight volatility is quite distinct from intraday volatility, implying that it should be modeled as an exogenous component.

We refer to the proposed approach as the multiple intraday HAR (miHAR) model in the subsequent empirical analysis. The stationarity conditions of this model are non-trivial and require a more in-depth analysis than the scope of this paper permits.

2.3. Benchmarks

Three benchmark models are chosen to compare with the miHAR. The first model is the HAR with diurnal effects (here denoted as diHAR) considered by Zhang et al. (2023). The diHAR model is given by

$$OK_{i,t} = \alpha + \beta^{\tau}D_i + \beta^{s}OK_{i-1,t} + \beta^{d}OK^{d}_{i,t-1} + \beta^{w}OK^{w}_{i,t-1} + \beta^{m}OK^{m}_{i,t-1} + \varepsilon_{i,t} \quad (4)$$

where D_i is the average diurnal volatility for the intraday interval i over the last 21 days. The diHAR model can be viewed as a severely constrained version of the proposed miHAR model. While the diHAR is designed to deal with the diurnal effect in intraday volatility by including D_i, its parameters on past intraday and daily volatility are not interval-specific. In contrast, the miHAR specification allows all parameters to vary over the trading day.

It is well known that volatility is time varying and undergoes structural breaks. Hence, non-linear models may be beneficial in this context. A popular way of dealing with non-linearity is to use tree-based methods (Hastie, Tibshirani, & Friedman, 2009). With this in mind, our second benchmark model is LightGBM (in the results, denoted as liGBM). LightGBM, short for the light gradient boosting machine and originally developed by Microsoft, is designed to be faster and more efficient than the popular XGBoost algorithm. LightGBM has performed

well in forecasting competitions such as the M5 competition (Makridakis, Spiliotis, & Assimakopoulos, 2022). As mentioned above, LightGBM is comparatively fast and computationally efficient, and given the size of the series used here, avoiding undue computational burden is an important issue. To be consistent with the data used as inputs for the miHAR and diHAR models, the spot volatility over the past 22 days is used as input for LightGBM. As discussed below in more detail, this equates to 1716 inputs with rolling estimation windows of 1000 trading days.

The third and final benchmark model is the deep-learning long short-term memory (LSTM) neural network. The LSTM architecture, developed by Hochreiter and Schmidhuber (1997), is an extension of the recurrent neural network (RNN), specifically designed to capture long-term dependencies. In contrast to a standard RNN architecture, LSTM contains a hidden state to store past information. This helps capture the long-term dependencies in the data—an important issue here, given the strong persistence in volatility series. LSTMs have been applied to forecast lower-frequency volatility, for example, by Bucci (2020). Zhang et al. (2023) also employed LSTM as one of the competing models for forecasting intraday volatility.

2.4. Performance measures

Following the literature on volatility forecast comparison (Patton, 2011; Patton & Sheppard, 2009), we use the empirical mean squared error (MSE) and quasi-likelihood (QLIKE) to assess out-of-sample forecast accuracy. For the basic case with one-step-ahead forecasts, these performance measures are defined as

$$MSE = \frac{1}{MT}\sum_{i=1}^{M}\sum_{t=1}^{T}\left(OK_{i,t} - F_{i,t}\right)^2, \tag{5}$$

and

$$QLIKE = \frac{1}{MT}\sum_{i=1}^{M}\sum_{t=1}^{T}\left(\frac{OK_{i,t}}{F_{i,t}} - \log\frac{OK_{i,t}}{F_{i,t}} - 1\right), \tag{6}$$

where T is the (common) number of forecasts for the intraday interval i, and $F_{i,t}$ denotes a one-step-ahead forecast of $OK_{i,t}$ from the miHAR model or one of the benchmark models at time $t-1$. Throughout, log is used to denote the natural logarithm.

Statistically significant differences in forecast performance are assessed using the model confidence set (MCS) of Hansen, Lunde, and Nason (2011). The MCS procedure starts with a collection of competing models, $\mathcal{M}^0$, indexed by $i = 1, \ldots, m_0$. Given a loss function, loss differentials $d_{ij,t}$ between models i and j are computed, and $H_0 : E(d_{ij,t}) = 0$ for all i, j (the null hypothesis of equal predictive accuracy, EPA) is tested. If the null hypothesis is rejected at significance level α, the worst-performing model is eliminated and the process is repeated until non-rejection occurs, with the set of surviving models being the MCS. By using the same $0 < \alpha < 1$ for all tests, the MCS contains the best model(s) from $\mathcal{M}^0$ with a limiting $(1-\alpha)100\%$ level of confidence. In our empirical study, the tests for EPA use the range statistic described by Hansen, Lunde, and Nason (2003).[1]

3. Empirical results

3.1. Data

For each of the six US blue-chip stocks in our study, we computed spot volatility estimates based on five-minute intraday intervals using Eq. (1), resulting in $M = 78$ intervals per trading day. Open, high, low, and close prices for each five-minute interval were sourced from LSEG DataScope Select. Following Zhang et al. (2023), we winsorized the volatility estimates for each stock at the 0.05th and 99.95th percentiles to mitigate the impact of outliers.

Although out-of-sample forecasting results are presented for all six stocks, the main discussion of the empirical results focuses on IBM. The sample period spans 3 January 2000 to 29 June 2023, encompassing a total of 5766 trading days or 449,748 five-minute spot volatility estimates.

3.2. In-sample results

While the focus of this paper is on the practical application of the computationally inexpensive miHAR model for forecasting, this section presents various in-sample estimation results to highlight important features of the model. We first examine its ordinary least squares (OLS) parameter estimates to gain insights into the intraday behavior of spot volatility. Following that, we present results demonstrating the miHAR model's ability, along with that of the competing benchmark models, to capture the dynamics of spot volatility. These results are based on the full IBM sample.

Fig. 1 shows the estimated parameters (and associated HAC confidence intervals) of the miHAR model across the $M = 78$ intraday intervals. The top panel presents the estimated intercepts, $\hat{\beta}_{i,0}$ $(i = 1, \ldots, M)$. These reflect the well-known diurnal pattern in volatility, with OLS estimates of $\beta_{i,0}$ being highest in the early part of the day, lowest during the middle (with parameters often not significantly different from zero), and rising again late in the trading day. To examine the effect of the overnight volatility during the trading day, the middle panel presents the estimated parameters on the overnight volatility, $\hat{\beta}_{i,4}$. It appears that overnight volatility has an impact during the first hour in the trading day but little effect after that, with most parameters not being significantly different from zero. To consider the effect of past (interval-specific) daily volatility and past intraday volatility, the bottom panel presents the sum of the estimated parameters on the spot volatility proxies for the previous day, week, and month, $\hat{\beta}_{i,1} + \hat{\beta}_{i,2} + \hat{\beta}_{i,3}$ (solid circle), and the sum of the estimated parameters on the averages of past proxied intraday volatility, $\hat{\beta}_{i,5} + \hat{\beta}_{i,6} + \hat{\beta}_{i,7}$ (solid triangle). During the onset of the trading day, volatility from past days exerts

[1] The MCS results reported in the study were obtained using code from the MFE toolbox of Kevin Sheppard, https://www.kevinsheppard.com/MFE_Toolbox.

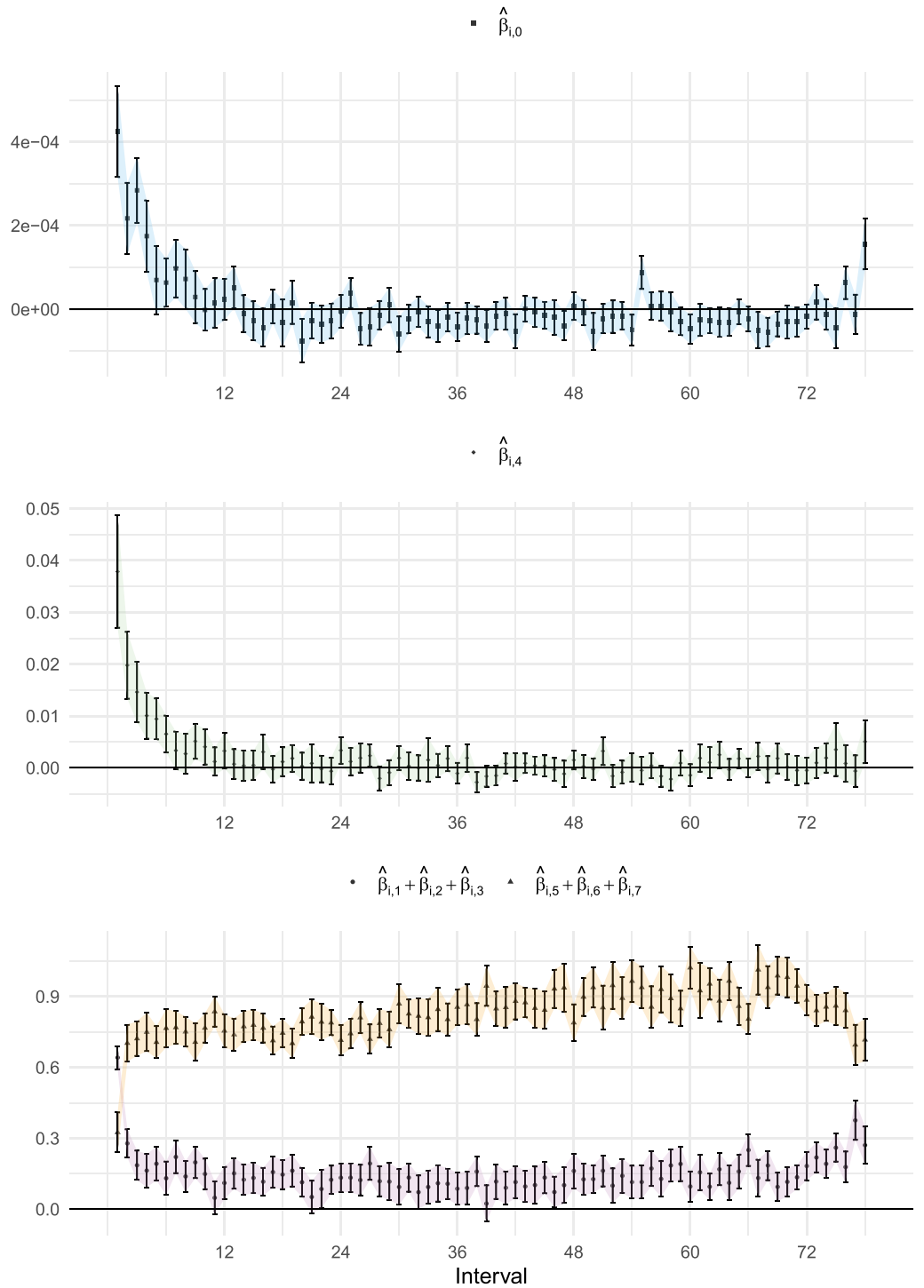


Fig. 1. OLS parameter estimates with approximate 95% HAC confidence intervals for the miHAR model using the IBM data. The horizontal axis represents the $M = 78$ five-minute intervals during the trading day. The top panel shows the estimated intercepts, $\hat{\beta}_{i,0}$ $(i = 1, \ldots, M)$. The middle panel shows the estimated parameters on the overnight volatility, $\hat{\beta}_{i,4}$. The bottom panel shows the sum of the estimated parameters on the spot volatility proxies for the previous day, week, and month $\hat{\beta}_{i,1} + \hat{\beta}_{i,2} + \hat{\beta}_{i,3}$ (solid circles), and the sum of the estimated parameters on the moving averages of past proxied intraday volatility, $\hat{\beta}_{i,5} + \hat{\beta}_{i,6} + \hat{\beta}_{i,7}$ (solid triangles).

a strong influence, quickly giving way to the dominance of recent intraday volatility as the day progresses. In the later part of the day, the impact of past intraday volatility weakens somewhat, with $\hat{\beta}_{i,5} + \hat{\beta}_{i,6} + \hat{\beta}_{i,7}$ (solid triangle) decreasing, while the influence of volatility from previous days slightly increases. These results provide new insights into the behavior of intraday volatility and highlight the challenge of capturing its dynamics with a model featuring fixed parameters throughout the trading day. A very similar pattern is observed for the other five stocks (see the online appendix). This indicates that estimation under commonality across different stocks may be beneficial.

To assess the effectiveness of the proposed miHAR model and competing benchmark models at capturing the dynamics of intraday spot volatility, proxied by the OK spot volatility estimator, Fig. 2 shows the sample autocorrelations (SACs) for the IBM $OK_{i,t}$ data, along with the SACs for the residuals of the miHAR and benchmark models. The top panel presents the SACs for the raw $OK_{i,t}$ data, illustrating its cyclical autocorrelations, indicative of the diurnal pattern. These autocorrelations decay very slowly across the 10 trading days (780 five-minute intervals), reflecting long-term persistence. The bottom panel presents the SACs for the residuals of the competing models. The SACs for the residuals of the diHAR model (hollow circles) indicate that the model captures only some of the dynamics in $OK_{i,t}$. While the overall level of autocorrelation in these residuals is lower than that of raw proxied volatility, substantial autocorrelations persist and a degree of cyclicality remains. The LSTM (solid triangles) performs better at capturing the dynamics. While a degree of cyclicality remains in the SACs of its residuals, the magnitude is smaller. Overall, miHAR (solid circles) and liGBM (hollow triangles) perform the best. The SACs of the residuals for both models have smaller magnitudes and exhibit less pronounced cyclicality. These results highlight the ability of the multiple-equation regression-based miHAR model to capture the complex dynamics of intraday spot volatility. The miHAR model performs similarly to or better than the more advanced, computationally intensive benchmark machine learning and deep learning models. The results also underscore the advantage of the added flexibility in the miHAR model compared to the fixed parameters across the trading day in the diHAR model. While miHAR performs well in this context, its out-of-sample forecasting performance, which is of central interest here, is addressed in the following section.

3.3. Out-of-sample results

We use a rolling window scheme to generate all out-of-sample forecasts. Following Clements and Preve (2021) and others, we use a 1000-day rolling window. The parameters of the regression-based miHAR and diHAR models are re-estimated each day, with forecasts generated for the following trading day. Following training on the initial 1000-day window, the LightGBM (liGBM) and LSTM models undergo retraining every 250 trading days to reduce computational cost. The subsequent empirical results are for six US blue-chip stocks, including IBM. The other stocks are American Express (AXP), Boeing (BA), 3M (MMM), Procter & Gamble (PG), and Walmart (WMT).

Table 1
One-step-ahead forecasting results for the miHAR and benchmark liGBM, LSTM, and diHAR intraday volatility models. MSE (top panel) and QLIKE (bottom panel) miHAR to miHAR, liGBM, LSTM, or diHAR performance measure ratios.

Symbol	miHAR	liGBM	LSTM	diHAR
		MSE		
AXP	1.000*	0.827	0.661	0.842
BA	1.000*	0.816	0.729	0.831
IBM	1.000*	0.875	0.723	0.847
MMM	1.000*	0.860	0.785	0.858
PG	1.000*	0.844	0.769	0.858
WMT	1.000*	0.890	0.787	0.842
		QLIKE		
AXP	1.000*	0.978	0.860	0.853
BA	1.000*	0.982	0.908	0.886
IBM	1.000*	0.976	0.909	0.877
MMM	1.000*	0.984	0.932	0.888
PG	1.000*	1.003*	0.949	0.913
WMT	1.000*	0.990	0.949	0.892

Asterisks highlight models included in a 90% MCS.

Table 1 reports our main forecasting results, comparing the proposed miHAR to liGBM, LSTM, and diHAR. The results are presented in the form of miHAR-to-benchmark (liGBM, LSTM, or diHAR) performance measure ratios. Values less than 1 indicate that the benchmark performs worse than miHAR. As revealed by both the MSE and QLIKE ratios, the miHAR model nearly always outperforms the three benchmark models in terms of one-step-ahead forecasts. The preceding statement is further evidenced by the MCS tests, one for each stock and performance measure, suggesting that the miHAR model is always in the set of superior models.

More specifically, the results reported in Table 1 show that with the exception of one case (PG under QLIKE loss), forecasts from the proposed miHAR have a lower MSE and QLIKE across all stocks. The improvements under the MSE are around 15% on average, relative to the benchmark liGBM and diHAR models, and more than 20% relative to the LSTM.[2] Under QLIKE loss, liGBM is closer in performance to miHAR than the diHAR and LSTM models. In the case of PG, miHAR and liGBM provide equivalent performance in terms of QLIKE. The MCS results indicate that miHAR is the only member of the MCS, with the exception of liGBM for PG under QLIKE. Overall, these results provide strong evidence that the forecasting performance of miHAR is superior to that of the popular tree-based machine learning method liGBM, the deep learning LSTM neural network, and the regression-based diHAR. Compared to the diHAR, the additional parameter flexibility

[2] Zhang et al. (2023) investigated the issue of commonality by training models on volatility data from a cross-section of stocks. They found that LSTM benefits from being trained on volatility from multiple assets. In unreported results, training an LSTM on all six stocks improved forecast performance, but the resulting forecasts remained inferior to those from the miHAR model and were excluded from the MCS. The issue of incorporating commonality into the miHAR and benchmark models is beyond the scope of the current paper and would require more extensive data than the current framework.

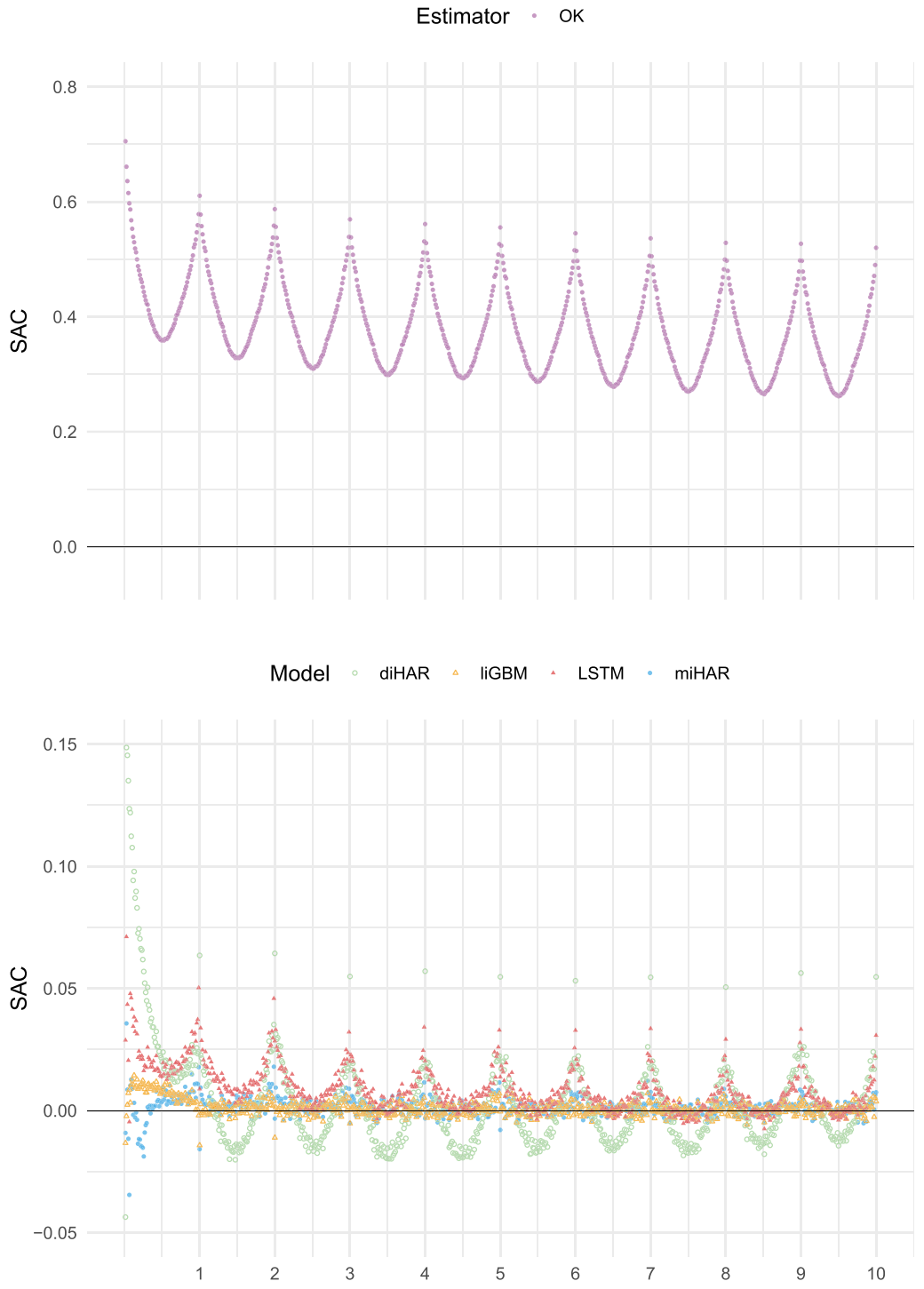


Fig. 2. Sample autocorrelations for lags up to 10 trading days (780 five-minute intervals) using the IBM data. Top panel: raw $OK_{i,t}$. Bottom panel: model residuals—miHAR (solid circles), diHAR (hollow circles), liGBM (hollow triangles), and LSTM (solid triangles).

offered by miHAR seemingly leads to superior forecasting performance.

To gain a deeper understanding of the performance of miHAR, we examined the interval-specific loss:

$$MSE_i = \frac{1}{T}\sum_{t=1}^{T}\left(OK_{i,t} - F_{i,t}\right)^2, \tag{7}$$

and

$$QLIKE_i = \frac{1}{T}\sum_{t=1}^{T}\left(\frac{OK_{i,t}}{F_{i,t}} - \log\frac{OK_{i,t}}{F_{i,t}} - 1\right). \tag{8}$$

Fig. 3 plots the ratios of the average miHAR loss to the average liGBM loss (as the best-performing benchmark) for each intraday interval, $i = 1, \ldots, M$. Following the results in Table 1, ratios less than one indicate that miHAR provides superior performance relative to liGBM. The top panel shows results for the MSE, while the bottom panel presents results for QLIKE. The MSE results show that miHAR outperforms liGBM by a wide margin in the early part of the trading day, and then consistently across all subsequent intervals. A broadly similar pattern is observed under QLIKE in the bottom panel, though miHAR underperforms liGBM in a few intervals later in the day. Overall, the main improvements in both MSE and QLIKE appear at the start and end of the trading day, as indicated by the superimposed smooth curves fitted using the LOESS method. While not reported here, similar results are observed when comparing miHAR to LSTM and diHAR. These results extend those reported in Table 1 to show that miHAR consistently outperforms the competing models throughout the trading day.

3.4. Robustness checks

This section presents results that examine the robustness of the benefits of the miHAR framework across various dimensions. These dimensions include logarithmic extensions, multi-step-ahead forecasts, the impact of overnight volatility, the role of the diurnal pattern, and simpler model specifications.

3.4.1. Logarithmic models

Models based on the (natural) logarithm of realized volatility at a daily frequency have often been found to provide superior forecasting performance. Andersen, Bollerslev, Diebold, and Labys (2003) were among the first to apply the logarithmic transformation, which has since become a standard benchmark for HAR models at a daily frequency. In our case, $\log(OK_{i,t})$ is the proxy for latent logarithmic intraday volatility used for estimation and training purposes. Denote by $\widetilde{F}_{i,t}$ a one-step-ahead forecast of $\log(OK_{i,t})$ from one of the competing models at time $t-1$. A bias-corrected forecast of intraday volatility in levels is then given by

$$F_{i,t} = \exp\left(\widetilde{F}_{i,t} + \frac{\sigma^2}{2}\right), \tag{9}$$

where, following Proietti and Lütkepohl (2013), we assume the one-step-ahead prediction error variance σ^2 to be interval- and time-invariant. In practice, we replace σ^2 in Eq. (9) with the sample variance of the residuals from the rolling window scheme.

Table 2
One-step-ahead forecasting results for log-miHAR and for the competing log-liGBM, log-LSTM, and log-diHAR intraday volatility models. MSE (top panel) and QLIKE (bottom panel) log-miHAR to log-miHAR, log-liGBM, log-LSTM, or log-diHAR performance measure ratios.

Symbol	log-miHAR	log-liGBM	log-LSTM	log-diHAR
		MSE		
AXP	1.000*	0.674	0.609	0.862
BA	1.000*	0.658	0.663	0.810
IBM	1.000*	0.675	0.664	0.837
MMM	1.000*	0.590	0.733	0.839
PG	1.000*	0.719	0.692	0.835
WMT	1.000*	0.724	0.738	0.833
		QLIKE		
AXP	1.000*	0.807	0.879	0.891
BA	1.000*	0.817	0.901	0.885
IBM	1.000*	0.788	0.895	0.885
MMM	1.000*	0.766	0.918	0.892
PG	1.000*	0.826	0.913	0.903
WMT	1.000*	0.810	0.738	0.894

Asterisks highlight models included in a 90% MCS.

Table 2 presents the forecasting results for the miHAR, liGBM, LSTM, and diHAR models applied to the logarithm of OK. We denote these logarithmic extensions by log-miHAR, log-liGBM, log-LSTM, and log-diHAR, respectively. The extension of the miHAR model continues to dominate and is now the only model included in the MCS. The difference in performance between the log-miHAR, log-LSTM, and log-diHAR models in terms of relative measures is similar to the above results in Table 1. An interesting result is that the performance of log-liGBM and log-LSTM deteriorates compared to the two regression-based logarithmic approaches. This suggests that the regression-based methods benefit more from the logarithmic transformation than the machine learning and deep learning methods. In conclusion, there is strong evidence that the benefits of the proposed miHAR are robust to the use of logarithmic OK.

3.4.2. Multi-step-ahead forecasts

The results presented thus far are based on one-step-ahead volatility forecasts over the next five-minute interval. We next consider the benefits of miHAR in the context of multi-step-ahead forecasts: two and six steps ahead, that is, for the next 10- and 30-minute intervals. Following Bollerslev, Patton, and Quaedvlieg (2016), our focus is on direct forecasting, where the one-period volatility estimate on the left-hand sides of Eqs. (2), (3), and (4) is replaced by the average of $OK_{i,t}$ over the forecast horizon. This average becomes the target variable for estimating the miHAR and diHAR models and training the liGBM and LSTM models. The predictor variables remain the same when multi-step-ahead forecasts are considered, but their estimated parameters change due to the new target variable. This follows the standard approach used for HAR models at a daily frequency, as in Bollerslev et al. (2016) and Clements and Preve (2021), among many others. An

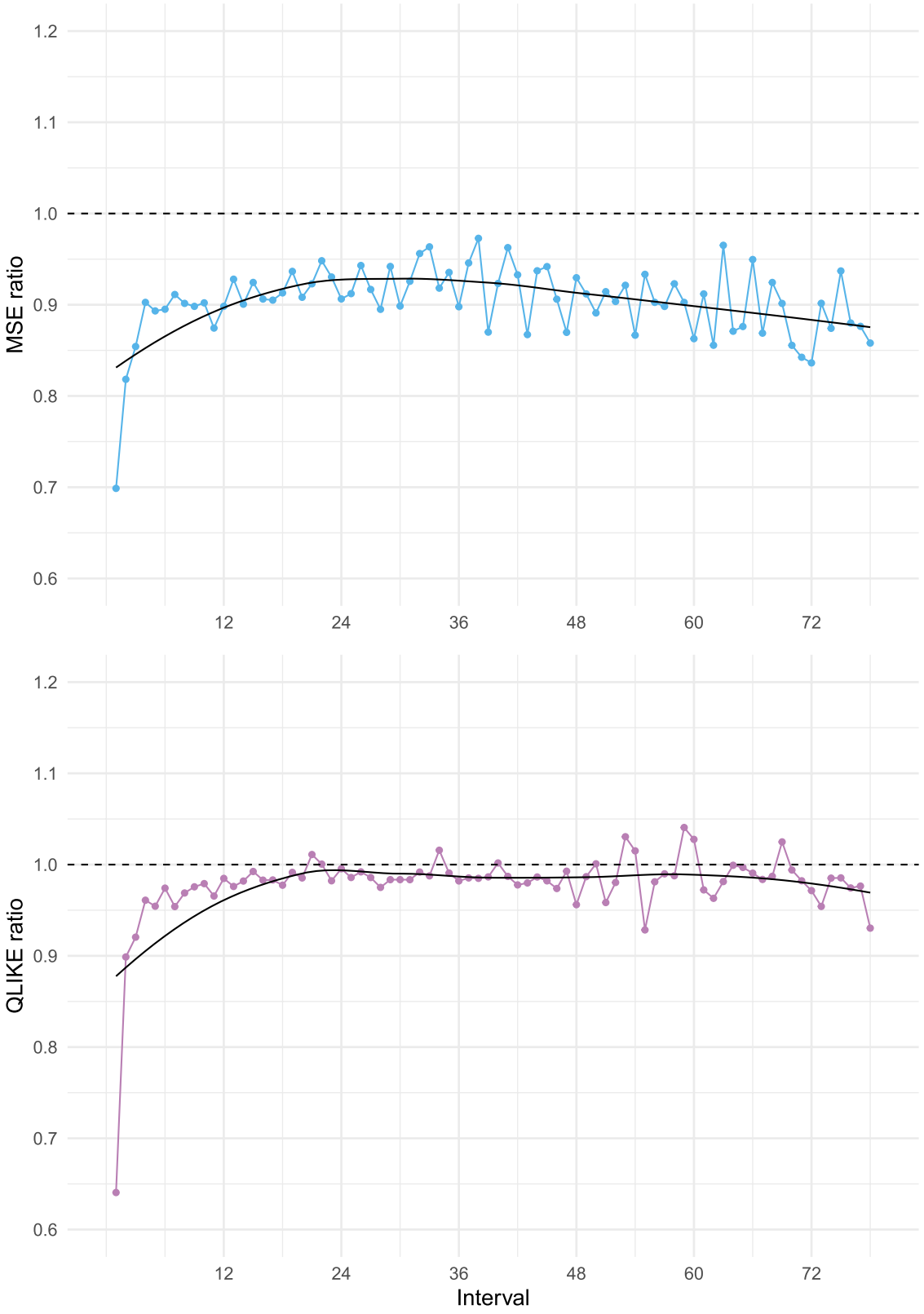


Fig. 3. Ratios of the average miHAR loss to liGBM loss for each intraday interval ($i = 1, \ldots, M$) using IBM data. Top panel: MSE results. Bottom panel: QLIKE results. Smooth curves, fitted using the LOESS method, are superimposed.

Table 3
Multi-step-ahead forecasting results for the miHAR and benchmark liGBM, LSTM, and diHAR intraday volatility models. MSE (top panel) and QLIKE (bottom panel) miHAR to miHAR, liGBM, LSTM, or diHAR performance measure ratios.

Symbol	Two steps ahead				Six steps ahead			
	miHAR	liGBM	LSTM	diHAR	miHAR	liGBM	LSTM	diHAR
Panel A: $OK_{i,t}$								
				MSE				
AXP	1.000*	0.780	0.647	0.777	1.000*	0.626	0.564	0.715
BA	1.000*	0.765	0.742	0.766	1.000*	0.690	0.675	0.695
IBM	1.000*	0.844	0.667	0.783	1.000*	0.730	0.544	0.701
MMM	1.000*	0.831	0.765	0.794	1.000*	0.781	0.681	0.714
PG	1.000*	0.797	0.765	0.796	1.000*	0.672	0.645	0.710
WMT	1.000*	0.851	0.808	0.779	1.000*	0.770	0.737	0.716
				QLIKE				
AXP	1.000*	0.947	0.955	0.789	1.000*	0.873	0.829	0.721
BA	1.000	0.962	1.051*	0.824	1.000	0.920	1.160*	0.753
IBM	1.000*	0.963	0.964	0.819	1.000*	0.909	0.993*	0.742
MMM	1.000	0.966	1.016*	0.826	1.000	0.926	1.045*	0.751
PG	1.000	0.972	1.042*	0.847	1.000	0.935	1.083*	0.783
WMT	1.000	0.975	1.069*	0.837	1.000	0.931	1.108*	0.765
Panel B: $\log(OK_{i,t})$								
				MSE				
AXP	1.000*	0.747	0.610	0.846	1.000*	0.590	0.576	0.762
BA	1.000*	0.761	0.710	0.800	1.000*	0.623	0.586	0.730
IBM	1.000*	0.802	0.660	0.809	1.000*	0.680	0.530	0.718
MMM	1.000*	0.852	0.740	0.821	1.000*	0.713	0.697	0.731
PG	1.000*	0.805	0.692	0.817	1.000*	0.684	0.671	0.727
WMT	1.000*	0.886	0.804	0.834	1.000*	0.797	0.774	0.762
				QLIKE				
AXP	1.000*	0.901	0.918	0.847	1.000*	0.824	0.984	0.743
BA	1.000*	0.920	0.963	0.849	1.000	0.850	1.033*	0.756
IBM	1.000*	0.930	0.907	0.844	1.000*	0.866	0.901	0.743
MMM	1.000*	0.928	0.954	0.855	1.000	0.863	1.069*	0.753
PG	1.000*	0.936	0.905	0.879	1.000*	0.873	0.995*	0.784
WMT	1.000*	0.949	0.943	0.867	1.000	0.882	1.041*	0.768

Asterisks highlight models included in a 90% MCS.

alternative approach would be to use iterative (or indirect) multi-step-ahead forecasts, which rely on iterating one-step-ahead forecasts. We prefer direct forecasts, as they are easier to compute and more robust to model misspecification (Chevillon & Hendry, 2005). Moreover, incorporating additional exogenous variables (such as order book data, liquidity, or trading volume) would make indirect forecasting even more challenging, as it would require forecasting models for each of these variables.

The same rolling window scheme discussed above is employed. Multi-step-ahead forecasts are generated by initiating the process at the beginning of each trading day. Consequently, the number of forecasts decreases daily as the forecast horizon expands (78 five-minute intervals - forecast horizon + 1).

Table 3 presents multi-step-ahead forecasting results for two- and six-step-ahead horizons. Panel A reports results for the miHAR, liGBM, LSTM, and diHAR models applied to raw proxied volatility, $OK_{i,t}$. Panel B presents the corresponding results for these models applied to logarithmic proxied volatility, $\log(OK_{i,t})$. The results in Panel A show that the miHAR model outperforms the three benchmark models also for the two- and six-step-ahead horizons under the MSE. However, under QLIKE, there are several instances where only LSTM is included in the MCS. The results in Panel B show that miHAR once again outperforms the benchmarks at all horizons under the MSE. Under QLIKE, for the two-step-ahead horizon, only miHAR is included in the MCS, while there are a few instances where only the LSTM is included for the six-step-ahead horizon. Although there are instances under QLIKE where LSTM outperforms miHAR for multi-step-ahead forecasts when applied to raw proxied volatility, when applied to logarithmic proxied volatility, these instances decrease. This provides further evidence that regression-based forecasts benefit from the logarithmic transformation.

3.4.3. Overnight volatility

We next investigate the robustness of the miHAR approach by excluding overnight volatility. More specifically, forecasts are generated using the miHAR model with the restriction $\beta_{i,4} = 0$ in Eqs. (2) and (3), effectively eliminating the effect of overnight volatility on the following trading day's forecasts. The model under the restriction $\beta_{i,4} = 0$ $(i = 1, \ldots, M)$ is henceforth referred to as miHAR*.

Before evaluating the forecasting performance of the miHAR* model, it is useful to first examine some in-sample estimation results to understand the impact of excluding overnight volatility. Fig. 4 presents the estimated parameters and corresponding HAC confidence intervals for the miHAR* (reduced miHAR) model, making it directly comparable to Fig. 1, which shows the results for

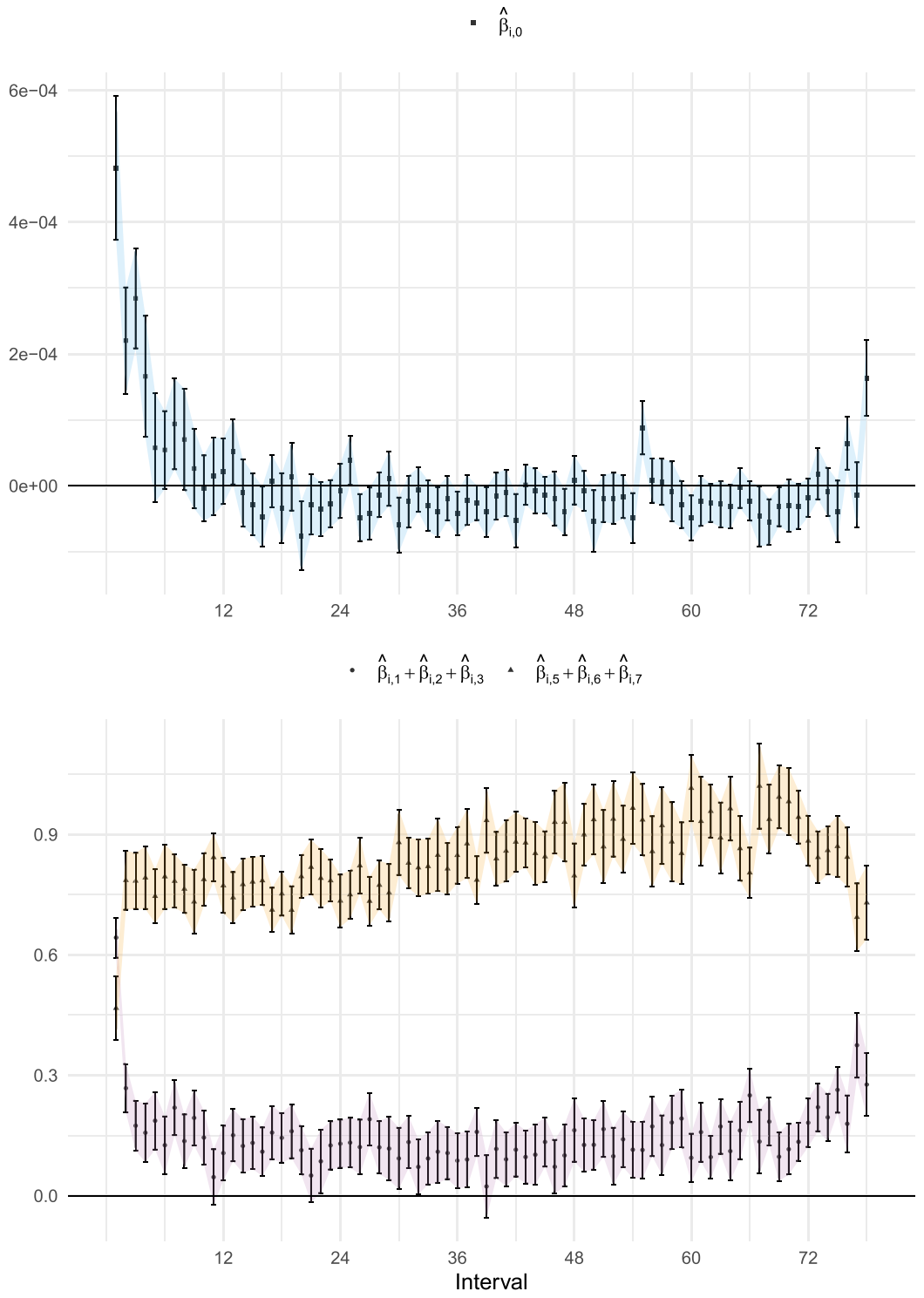


Fig. 4. OLS parameter estimates with approximate 95% HAC confidence intervals for the reduced miHAR model, excluding overnight volatility and using IBM data. The top panel shows the estimated intercepts $\hat{\beta}_{i,0}$ $(i = 1, \ldots, M)$. The bottom panel shows the sum of the estimated parameters on the spot volatility proxies for the previous day, week, and month $\hat{\beta}_{i,1} + \hat{\beta}_{i,2} + \hat{\beta}_{i,3}$ (solid circles), and the sum of the estimated parameters on the moving averages of past proxied intraday volatility, $\hat{\beta}_{i,5} + \hat{\beta}_{i,6} + \hat{\beta}_{i,7}$ (solid triangles).

Table 4
Forecasting results when overnight volatility is excluded from the miHAR model ($\beta_{i,4} = 0$ in Eqs. (2) and (3)). One-step-ahead results for the miHAR* and benchmark liGBM, LSTM, and diHAR intraday volatility models. MSE (top panel) and QLIKE (bottom panel) miHAR* to miHAR*, liGBM, LSTM, or diHAR performance measure ratios.

Symbol	miHAR*	liGBM	LSTM	diHAR
		MSE		
AXP	1.000*	0.838	0.669	0.853
BA	1.000*	0.838	0.749	0.854
IBM	1.000*	0.904	0.746	0.875
MMM	1.000*	0.879	0.801	0.876
PG	1.000*	0.864	0.786	0.877
WMT	1.000*	0.919	0.812	0.869
		QLIKE		
AXP	1.000*	0.984	0.864	0.857
BA	1.000*	0.989	0.914	0.892
IBM	1.000*	0.982	0.914	0.882
MMM	1.000*	0.990	0.939	0.894
PG	1.000*	0.994	0.940	0.905
WMT	1.000*	0.998*	0.956	0.899

Asterisks highlight models included in a 90% MCS.

the miHAR model. The top panel in Fig. 4 displays the estimated intercepts, $\hat{\beta}_{i,0}$ ($i = 1, \ldots, 78$), which closely resemble those reported in Fig. 1. The only notable difference is the larger value of $\hat{\beta}_{1,0}$ in the first interval. The bottom panel displays the sum of the estimated parameters on the spot volatility proxies for the previous day, week, and month, $\hat{\beta}_{i,1} + \hat{\beta}_{i,2} + \hat{\beta}_{i,3}$ (solid circles), along with the sum of the estimated parameters on the moving averages of past proxied intraday volatility, $\hat{\beta}_{i,5} + \hat{\beta}_{i,6} + \hat{\beta}_{i,7}$ (solid triangles). Compared to the corresponding panel in Fig. 1, there are few discernible differences. These results indicate that the overall model structure remains robust despite the exclusion of overnight volatility. Next, we assess whether this exclusion harms forecasting performance.

Table 4 presents the forecasting results for the miHAR* model. Overall, while there are some small reductions in relative forecast performance when overnight volatility is excluded from the miHAR (for instance, the improvements under MSE reduce to around 13% on average), the results demonstrate that miHAR* still outperforms the benchmark models in nearly all cases, with the exception of WMT under QLIKE loss. This suggests that the superior forecasts provided by the proposed multiple-equation regression-based approach are not solely attributable to the inclusion of overnight volatility. The enhanced parameter flexibility compared to the diHAR model proves beneficial at improving forecast performance, indicating the significance of time-of-day-dependent dynamics. The proposed miHAR achieves this in a parsimonious manner.

3.4.4. The diurnal pattern

Christensen, Hounyo, and Podolskij (2018) examine whether the diurnal pattern alone is sufficient to explain the intraday dynamics in volatility. While they find that accounting for the diurnal pattern does capture a substantial portion of these dynamics, a significant structure persists in intraday returns even after controlling for the diurnal volatility pattern. These findings are purely explanatory and do not pertain to forecasting. Therefore, we aim to determine whether the benefits of the proposed miHAR model come solely from capturing the diurnal pattern. In other words, does ignoring the intraday dynamics for spot volatility incur any costs for forecasting?

In this study, our approach involves scaling the proxied spot volatility using the estimated diurnal pattern and incorporating longer-term daily effects. We then proceed to forecast these daily effects before finally rescaling them by the fixed diurnal pattern. To this end, let

$$\widetilde{OK}_{i,t} = \frac{OK_{i,t}}{\overline{OK}_i} \tag{10}$$

denote the scaled OK series for some asset, where $\overline{OK}_i$ is the average proxied spot volatility for intraday interval $i = 1, \ldots, M$. We employ modified versions of Eqs. (2) and (3), devoid of intraday dynamics and henceforth referred to as miHAR**, to capture the longer-term daily persistence in scaled proxied spot volatility

$$\widetilde{OK}_{i,t} = \beta_{i,0} + \beta_{i,1}\widetilde{OK}^{d}_{i,t-1} + \beta_{i,2}\widetilde{OK}^{w}_{i,t-1} + \beta_{i,3}\widetilde{OK}^{m}_{i,t-1} + \varepsilon_{i,t}. \tag{11}$$

Forecasts of scaled spot volatility are first obtained from Eq. (11). A (rescaled) forecast of spot volatility in levels is then given by

$$F_{i,t} = \widetilde{F}_{i,t}\overline{OK}_i, \tag{12}$$

where $\widetilde{F}_{i,t}$ denotes a one-step-ahead forecast of scaled spot volatility.

Alternative approaches to capturing the diurnal pattern, beyond scaling by $\overline{OK}_i$, could be considered. One option is a smooth approximation to $\overline{OK}_i$, such as the flexible Fourier form used by Andersen and Bollerslev (1997). However, while OLS estimates will differ numerically between models with differently scaled variables, the forecasts will be identical once the scaling transformations for the target variable are correctly reversed.

Table 5 presents the forecasting results comparing the miHAR** to the original miHAR, and to the liGBM and LSTM. It is clear that the miHAR** forecasts, accounting for the diurnal pattern (Eq. (10)) and longer-term daily effects (Eq. (11)), obtained using Eq. (12), are inferior to the miHAR and to the liGBM and LSTM models. These results indicate that solely capturing the diurnal pattern along with the daily effects is not sufficient for accurate forecasting. In other words, ignoring intraday dynamics comes at a cost in terms of forecast performance. Additionally, the performance measure ratios for the miHAR** are considerably smaller than those reported in Table 1 for diHAR, suggesting that it is preferable to incorporate even restrictive intraday dynamics, as captured in the diHAR model, rather than relying solely on the diurnal effect. These findings extend those of Christensen et al. (2018) to the forecasting context, confirming that while the diurnal pattern in volatility is important, further intraday dynamics must be addressed for accurate forecasting.

Table 5
One-step-ahead forecasting results for the miHAR and alternative liGBM, LSTM, and miHAR** intraday volatility models. MSE (top panel) and QLIKE (bottom panel) miHAR to miHAR, liGBM, or miHAR** performance measure ratios.

Symbol	miHAR	liGBM	LSTM	miHAR**
		MSE		
AXP	1.000*	0.827	0.661	0.663
BA	1.000*	0.816	0.730	0.642
IBM	1.000*	0.875	0.723	0.656
MMM	1.000*	0.860	0.785	0.674
PG	1.000*	0.844	0.769	0.670
WMT	1.000*	0.890	0.723	0.662
		QLIKE		
AXP	1.000*	0.980	0.860	0.707
BA	1.000*	0.983	0.908	0.692
IBM	1.000*	0.976	0.909	0.695
MMM	1.000*	0.985	0.932	0.716
PG	1.000*	1.003*	0.949	0.743
WMT	1.000*	0.990	0.909	0.713

Asterisks highlight models included in a 90% MCS.

3.4.5. Simpler models

In this section, we examine the forecasting performance of various miHAR model variations, most of which are simpler than the standard model given by Eqs. (2) and (3). First, we explore the potential benefits of using the least absolute shrinkage and selection operator (LASSO). We next examine a variation with a single fixed equation, along with three smaller variations, for the mid-to-late part of the trading day.

We start with an approach similar to that of Audrino and Knaus (2016), where we apply the LASSO shrinkage method to an extension of the miHAR model without parameter constraints. We consider the full set of predictor variables as implied by Eqs. (2) and (3), using the LASSO for variable selection. This allows us to assess whether a smaller, more parsimonious model (with a reduced set of predictor variables) improves forecasting performance. The largest model includes 44 predictor variables, consisting of 22 daily lags of $OK_{i,t}$, 20 intraday lags of $OK_{i,t}$, the overnight volatility $|r_t^{ON}|$, and the intercept.

The unconstrained version of Eq. (2) can be written as

$$OK_{1,t} = \mathbf{X}_{1,t}\boldsymbol{\theta}_{1,t} + \varepsilon_{1,t}, \tag{13}$$

where $\boldsymbol{\theta}_{1,t}$ is a 44×1 parameter vector, and $\mathbf{X}_{1,t}$ is the 1×44 vector given by

$$\mathbf{X}_{1,t} = \big(1, OK_{1,t-1}, \ldots, OK_{1,t-22}, |r_t^{ON}|, OK_{M,t-1}, \ldots, OK_{M-19,t-1}\big). \tag{14}$$

Similarly, the unconstrained version of Eq. (3) can be written as

$$OK_{i,t} = \mathbf{X}_{i,t}\boldsymbol{\theta}_{i,t} + \varepsilon_{i,t}, \tag{15}$$

with $\mathbf{X}_{i,t}$ given by

$$\mathbf{X}_{i,t} = \big(1, OK_{i,t-1}, \ldots, OK_{i,t-22}, |r_t^{ON}|, OK_{i-1,t}, \ldots, OK_{i-20,t}\big). \tag{16}$$

The parameters $\boldsymbol{\theta}_{1,t}$ in Eq. (13) and $\boldsymbol{\theta}_{i,t}$ $(i = 2, \ldots, M)$ in Eq. (15) will be estimated using the LASSO. Variable selection using the LASSO may lead to a more accurate and interpretable model, as any elements in $\widehat{\boldsymbol{\theta}}1, t$ and $\widehat{\boldsymbol{\theta}}i, t$ that are shrunk to zero indicate that the corresponding predictors in $\mathbf{X}1, t$ and $\mathbf{X}i, t$ are irrelevant for forecasting $OK_{i,t}$.

To tune the LASSO hyperparameter, λ, we use the hv-block cross-validation method of Racine (2000). For a given sample of n observations on the target and predictors, a validation set consists of $n_v = 2v + 1$ observations centered on the ith observation. An additional h observations are removed on either side of the validation set, leaving $n_c = n - 2h - 2v - 1$ observations for a training set. While v controls the size of the validation set relative to the training set and the full sample, h is chosen to reduce dependence between the training and validation sets. The size of the training set, n_c, is chosen as the integer part of n^{δ}, where $\log p / \log n < \delta < 1$, and p is the number of predictors. In our rolling window scheme, we have $n = 978$ observations on the target and $p = 44$ predictors, based on a 1000-day window adjusted for a 22-day maximum lag. With $\delta = 0.95$ and $h = 50$, this yields a training set of 693 observations and a validation set of 185 observations.

To reduce the computational burden of cross-validation, we use five training–validation splits per window. For each split, the LASSO model is fitted on the training set, and its MSE is evaluated on the validation set over a grid of 100 λ values, logarithmically spaced and generated by MATLAB's built-in `lasso` function using the full sample of size n. For each λ value, we compute the average MSE across the five validation sets and select the λ that yields the lowest average MSE. The LASSO model is then re-estimated on the full 1000-observation window using this selected hyperparameter. Once trained, the model is ready for forecasting. Even with just five training–validation splits per window, this approach is computationally expensive under our rolling window scheme. This LASSO-based approach is henceforth referred to as miHAR^L.

Before evaluating the forecasting performance of the miHAR^L, it is useful to first examine some variable-selection results to understand the impact of applying the LASSO shrinkage method. Fig. 5 shows the number of daily predictors selected (top panel), whether the overnight volatility is included (middle panel), and the number of intraday predictors selected (bottom panel). The top panel reveals that, for most of the early sample, the majority of the 22 daily predictors are selected, irrespective of the intraday interval. This is followed by a period in which fewer predictors are selected. In most of the latter part of the sample, however, the majority of daily predictors are again selected. The middle panel shows that overnight volatility is often not selected during the early part of the sample, followed by a later period in which it is again frequently excluded during the second half of the trading day. For most of the sample, however, overnight volatility is selected regardless of the intraday interval. In contrast, the bottom panel shows that typically far fewer than all 20 intraday predictors are selected, with many instances in which few or none are chosen. In summary, among the sets of predictors (22 daily lags of $OK_{i,t}$, overnight volatility $|r_t^{ON}|$, and 20 intraday lags of

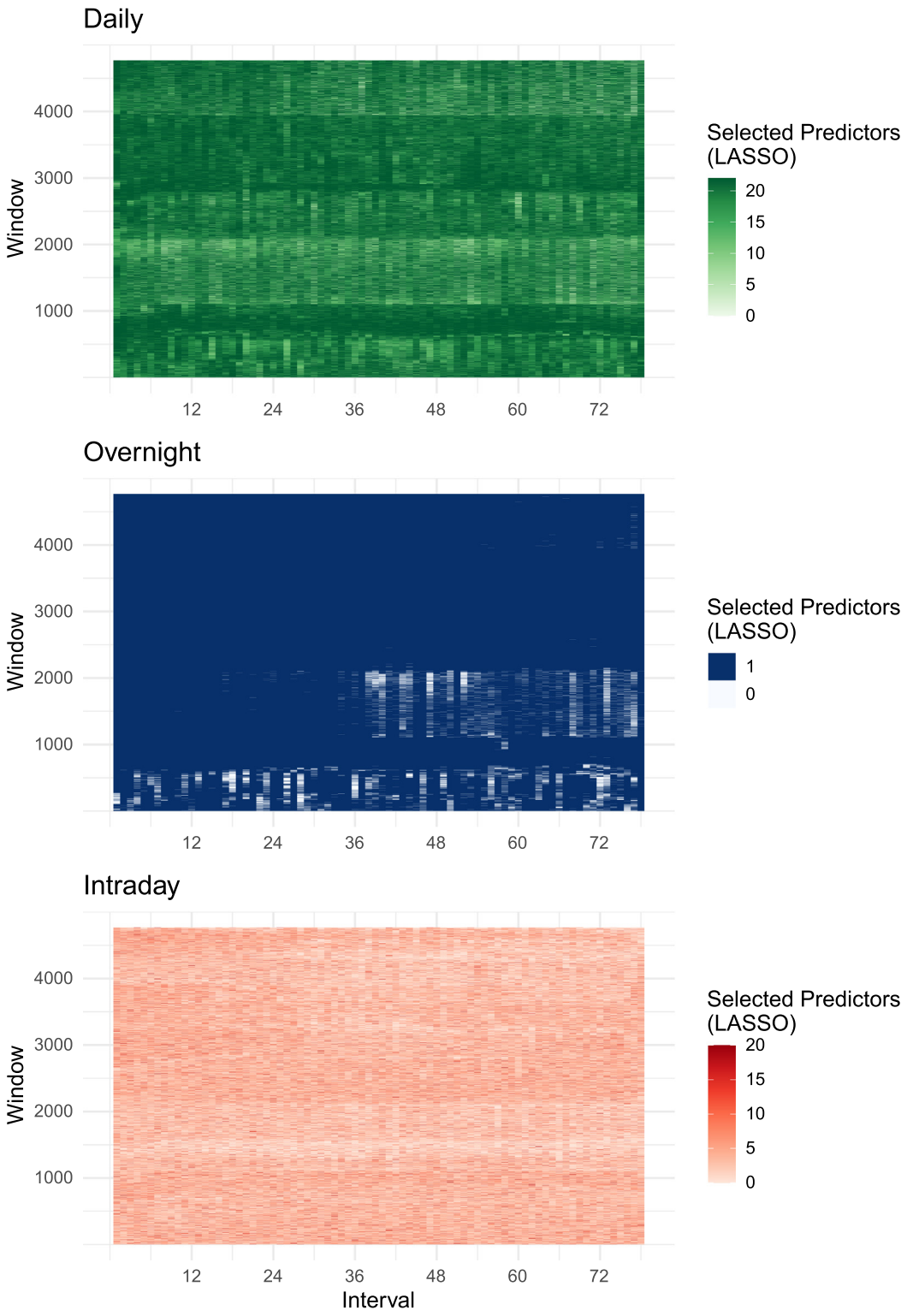


Fig. 5. Rolling window miHAR^{L} results with LASSO variable selection using IBM data, shown as heatmaps. Top panel: daily component (up to 22 predictors per interval). Middle panel: overnight component (up to one predictor per interval). Bottom panel: intraday component (up to 20 predictors per interval).

Table 6
One-step-ahead forecasting results for the miHAR^{L} and the standard miHAR, liGBM, LSTM, and diHAR intraday volatility models. MSE (top panel) and QLIKE (bottom panel) miHAR to miHAR, liGBM, LSTM, diHAR, or miHAR^{L} performance measure ratios.

Symbol	miHAR	liGBM	LSTM	diHAR	miHAR^{L}
			MSE		
AXP	1.000*	0.827	0.661	0.842	0.645
BA	1.000*	0.816	0.729	0.831	0.658
IBM	1.000*	0.875	0.723	0.847	0.653
MMM	1.000*	0.860	0.785	0.858	0.664
PG	1.000*	0.844	0.769	0.858	0.661
WMT	1.000*	0.890	0.787	0.842	0.662
			QLIKE		
AXP	1.000*	0.978	0.860	0.853	0.673
BA	1.000*	0.982	0.908	0.886	0.672
IBM	1.000*	0.976	0.909	0.877	0.662
MMM	1.000*	0.984	0.932	0.888	0.693
PG	1.000*	1.003*	0.949	0.913	0.714
WMT	1.000*	0.990	0.949	0.892	0.694

Asterisks highlight models included in a 90% MCS.

Table 7
One-step-ahead forecasting results for the miHAR^{F} and the standard miHAR, liGBM, LSTM, and diHAR intraday volatility models. MSE (top panel) and QLIKE (bottom panel) miHAR to miHAR, liGBM, LSTM, diHAR, or miHAR^{L} performance measure ratios.

Symbol	miHAR	liGBM	LSTM	diHAR	miHAR^{F}
			MSE		
AXP	1.000*	0.827	0.661	0.842	0.946
BA	1.000*	0.816	0.729	0.831	0.947
IBM	1.000*	0.875	0.723	0.847	0.919
MMM	1.000*	0.860	0.785	0.858	0.942
PG	1.000*	0.844	0.769	0.858	0.921
WMT	1.000*	0.890	0.787	0.842	0.936
			QLIKE		
AXP	1.000*	0.978	0.860	0.853	0.779
BA	1.000*	0.982	0.908	0.886	0.854
IBM	1.000*	0.976	0.909	0.877	0.863
MMM	1.000*	0.984	0.932	0.888	0.868
PG	1.000*	1.003*	0.949	0.913	0.815
WMT	1.000*	0.990	0.949	0.892	0.861

Asterisks highlight models included in a 90% MCS.

Table 8
One-step-ahead forecasting results for the standard miHAR and alternative miHAR^{DI}, miHAR^{OI}, and miHAR^{I} intraday volatility models. MSE (top panel) and QLIKE (bottom panel) miHAR to miHAR, miHAR^{DI}, miHAR^{OI}, or miHAR^{I} performance measure ratios.

Symbol	miHAR	miHAR^{DI}	miHAR^{OI}	miHAR^{I}
		MSE		
AXP	1.000	1.002*	1.000	1.002*
BA	1.000	1.002*	1.002	1.003*
IBM	1.000	1.001*	0.998	1.000
MMM	1.000	1.001*	0.998	0.999
PG	1.000	1.001*	1.000	1.001*
WMT	1.000	1.002*	0.998	0.999
		QLIKE		
AXP	1.000	1.002*	1.003	1.006*
BA	1.000	0.998	1.004	1.006*
IBM	1.000	1.002*	1.000	1.002*
MMM	1.000	1.001*	0.999	1.000*
PG	1.000	1.016	1.018	1.020*
WMT	1.000	1.002*	1.000	1.002*

Asterisks highlight models included in a 90% MCS.

$OK_{i,t}$), the LASSO frequently reduces the number of intraday lags and, during certain periods, nearly excludes them altogether. The impact of this on forecast performance is examined next.

Table 6 presents the forecasting results for the miHAR^{L}. The results from Table 1 are repeated here for comparison. It is evident that miHAR^{L} is inferior to the standard miHAR model and, in fact, performs worse than the benchmark models. These results highlight the importance of the miHAR intraday dynamics, as Fig. 5 shows that miHAR^{L} often reduces the number of intraday lags, resulting in inferior forecasting performance. This finding aligns with the results in Section 3.4.4, where it was shown that intraday dynamics (even when constrained) are important for forecasting accuracy.

The next variation of the miHAR model examined retains the full set of equations for the early part of the trading day while using a single fixed equation for the remainder of the day. A key decision is when to switch to the fixed equation, which can be framed as a hyperparameter tuning problem requiring a grid search (or a similar approach). For this robustness check, however, we adopt a simple and pragmatic approach. Given the OLS parameter estimates in Fig. 1, the miHAR parameters appear to stabilize after the first 15 intervals. Based on this, we use Eq. (2) for the first interval, instances of Eq. (3) for intervals $i = 2, \ldots, 15$, and a single fixed Eq. (3) for the remaining intervals $i = 16, \ldots, 78$. This approach is henceforth referred to as miHAR^{F}.

Table 7 presents the forecasting results for miHAR^{F}, with the results from Table 1 repeated for comparison. Under the MSE loss, miHAR^{F} outperforms the benchmark models but is inferior to the standard miHAR. Under the QLIKE loss, however, miHAR^{F} performs worse than all other models. These results highlight that the flexibility of the standard miHAR model throughout the entire trading day outweighs the risk of misspecification.

The final approach considered involves replacing a single fixed Eq. (3) for the mid-to-late part of the trading day ($i = 16, \ldots, 78$) with three smaller variations that use fewer predictors. The smaller variations correspond to omitting the overnight volatility, the daily lags, and both daily and overnight effects, while always including the intraday component. These variations of the standard miHAR are henceforth referred to as miHAR^{DI} (including daily and intraday effects but excluding overnight effects), miHAR^{OI} (including overnight and intraday effects but excluding daily effects), and miHAR^{I} (including only intraday effects, excluding both daily and overnight effects).

Table 8 presents the forecasting results for the simpler miHAR^{DI}, miHAR^{OI}, and miHAR^{I} alternatives to the standard miHAR. While the miHAR is excluded from the MCS in all cases, the differences in performance are very small (i.e. all ratios are close to one) compared to the previous results. Of the three simpler models, miHAR^{DI} and miHAR^{I}, which exclude the overnight effect and both daily and overnight effects, respectively, perform the best. Despite the similar forecasting performance of all four models, the large number of forecasts (4766 trading days with 78 intervals yields 371,748 forecasts using IBM data)

results in the MCS excluding some models. The OLS parameter estimates based on IBM data in Fig. 1 show that intraday predictors, unlike daily predictors, become increasingly important during the middle part of the trading day. A similar pattern is observed for the other five stocks (see the online appendix). These findings agree with the relative performance of miHAR^{DI} and miHAR^{I}, both of which include intraday predictors for this period, as well as with the LASSO-based forecasting results. Under LASSO, only a few intraday lags are typically selected, leading to a deterioration in forecasting performance and further reinforcing the importance of intraday predictors. The very small differences in forecasting performance (compared to the miHAR) suggest that the cost of changing the model structure during the day, whether based on a subjective choice or a computationally costly model-selection approach, does not seem to be justified.

4. Concluding remarks

In conclusion, this paper introduced a novel multiple-equation regression-based method for modeling and forecasting intraday spot volatility. Unlike conventional methods that treat the data as one continuous sample, our approach treats intraday intervals as individual time series with their own equations. By combining these equations, our framework captures diurnal, long-term daily, and intraday effects. The framework offers a straightforward approach for incorporating the effect of overnight volatility, is computationally inexpensive, and facilitates clear interpretation. Our empirical results demonstrated that the multiple-equation approach effectively captures the dynamics in intraday volatility and outperforms an alternative regression-based model (diHAR), a popular tree-based machine learning method (LightGBM), and a deep learning method (LSTM) in terms of forecasting performance. These improvements in predictive accuracy, as measured by the MSE and QLIKE, extend to logarithmic extensions and across multiple forecast horizons. Importantly, they are also robust to the exclusion of overnight volatility. The strong forecasting performance of the multiple-equation approach is not simply due to its ability to capture the diurnal pattern in volatility but rather reflects its capacity to model the more complex intraday dynamics. The standard multiple-equation model performs at least as well as several simpler variations that involve changing the model structure during the trading day.

Several interesting directions for future research emerge from this work. One possible avenue is estimating the multiple-equation model across multiple assets, which would require a more sophisticated regression framework that accounts for differences in parameters across assets. Another promising direction is a hybrid approach that combines regression with machine learning or deep learning methods. Given the well-understood structure in the data, regression could model these patterns, while machine learning or deep learning might be better suited for capturing more intricate intraday or daily effects. These ideas extend beyond the scope of this paper, which focused on presenting a framework that is easy to interpret, simple to implement, and computationally efficient.

CRediT authorship contribution statement

Adam Clements: Writing – review & editing, Writing – original draft, Software, Methodology, Data curation, Conceptualization. **Daniel P.A. Preve:** Writing – review & editing, Writing – original draft, Methodology, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A. Supplementary data

Supplementary material related to this article can be found online at https://doi.org/10.1016/j.ijforecast.2025.11.009.

Data and code availability

All data and code necessary to reproduce the results presented in this manuscript are available at https://github.com/AdamEClements/Intraday_Spot_Volatility_IJF_Replication/tree/main.

References

Andersen, T. G., & Bollerslev, T. (1997). Intraday periodicity and volatility persistence in financial markets. *Journal of Empirical Finance*, *4*, 115–158.

Andersen, T. G., Bollerslev, T., Diebold, F. X., & Labys, P. (2003). Modeling and forecasting realized volatility. *Econometrica*, *71*, 579–625.

Audrino, F., & Knaus, S. D. (2016). Lassoing the HAR model: A model selection perspective on realized volatility dynamics. *Econometric Reviews*, *35*(8–10), 1485–1521.

Bates, D. S. (2019). How crashes develop: Intradaily volatility and crash evolution. *The Journal of Finance*, *74*(1), 193–238.

Bollerslev, T., Patton, A. J., & Quaedvlieg, R. (2016). Exploiting the errors: A simple approach for improved volatility forecasting. *Journal of Econometrics*, *192*, 1–18.

Bucci, A. (2020). Realized volatility forecasting with neural networks. *Journal of Financial Econometrics*, *18*, 502–531.

Chevillon, G., & Hendry, D. F. (2005). Non-parametric direct multi-step estimation for forecasting economic processes. *International Journal of Forecasting*, *21*(2), 201–218.

Christensen, K., Hounyo, U., & Podolskij, M. (2018). Is the diurnal pattern sufficient to explain intraday variation in volatility? A nonparametric assessment. *Journal of Econometrics*, *205*(2), 336–362.

Clements, A., Hurn, S., & Li, Z. (2016). Forecasting day-ahead electricity load using a multiple equation time series approach. *European Journal of Operational Research*, *251*, 522–530.

Clements, A., & Preve, D. P. A (2021). A practical guide to harnessing the HAR volatility model. *Journal of Banking and Finance*, *133*, Article 106285.

Corsi, F. (2009). A simple approximate long-memory model of realized volatility. *Journal of Financial Econometrics*, *7*(2), 174–196.

Engle, R. F., & Sokalska, M. E. (2012). Forecasting intraday volatility in the US equity market. Multiplicative component GARCH. *Journal of Financial Econometrics*, *10*(1), 54–83.

Espinoza, M., Joye, C., Belmans, R., & De Moor, B. (2005). Short-term load forecasting, profile identification, and customer segmentation: A methodology based on periodic time series. *IEEE Transactions on Power Systems*, *20*, 1622–1630.

Hansen, P., Lunde, A., & Nason, J. (2003). Choosing the best volatility models: The model confidence set approach. *Oxford Bulletin of Economics and Statistics*, *65*, 839–861.

Hansen, P., Lunde, A., & Nason, J. (2011). The model confidence set. *Econometrica*, *79*, 453–497.

Hastie, T., Tibshirani, R., & Friedman, J. (2009). *The elements of statistical learning: Data mining, inference, and prediction*. Springer Science & Business Media.

Hochreiter, S., & Schmidhuber, Jürgen (1997). Long short-term memory. *Neural Computation*, *9*, 1735–1780.

Li, J., Wang, D., & Zhang, Q. (2024). Reading the candlesticks: An OK estimator for volatility. *The Review of Economics and Statistics*, *106*(4), 1114–1128.

Makridakis, S., Spiliotis, E., & Assimakopoulos, V. (2022). M5 accuracy competition: Results, findings, and conclusions. *International Journal of Forecasting*, *38*(4), 1346–1364.

Moisan, S., Herrera, R., & Clements, A. (2018). A dynamic multiple equation approach for forecasting $PM_{2.5}$ pollution in Santiago, Chile. *International Journal of Forecasting*, *34*, 566–581.

Patton, A. (2011). Volatility forecast comparison using imperfect volatility proxies. *Journal of Econometrics*, *160*, 246–256.

Patton, A. J., & Sheppard, K. (2009). Optimal combinations of realised volatility estimators. *International Journal of Forecasting*, *25*, 218–238.

Peirson, J., & Henley, A. (1994). Electricity load and temperature: Issues in dynamic specification. *Energy Economics*, *16*, 235–243.

Proietti, T., & Lütkepohl, H. (2013). Does the Box–Cox transformation help in forecasting macroeconomic time series? *International Journal of Forecasting*, *29*(1), 88–99.

Racine, J. (2000). Consistent cross-validatory model-selection for dependent data: *hv*-block cross-validation. *Journal of Econometrics*, *99*, 39–61.

Ramanathan, R., Engle, R., Granger, C. W. J., Vahid-Araghi, F., & Brace, C. (1997). Short-run forecasts of electricity loads and peaks. *International Journal of Forecasting*, *13*, 161–174.

Soares, L. J., & Medeiros, M. C. (2008). Modeling and forecasting short-term electricity load: A comparison of methods with an application to Brazilian data. *International Journal of Forecasting*, *24*, 630–644.

Stroud, J. R., & Johannes, M. S (2014). Bayesian modeling and forecasting of 24-hour high-frequency volatility. *Journal of the American Statistical Association*, *109*(508), 1368–1384.

Taylor, S. J., & Xu, X. (1997). The incremental volatility information in one million foreign exchange quotations. *Journal of Empirical Finance*, *4*(4), 317–340.

Zhang, C., Zhang, Y., Cucuringu, M., & Qian, Z. (2023). Volatility forecasting with machine learning and intraday commonality. *Journal of Financial Econometrics*, *22*(2), 492–530.